

Query 1 :

```
1 MATCH (m:Customer)-[:MADE_TRANSACTION]->(t:Terminal)<-
  [:MADE_TRANSACTION]-
2 (n:Customer)
3 WHERE m.id < n.id
4 WITH m,n,COUNT(DISTINCT t) AS sharedTerminals
5 WHERE sharedTerminals >= 4
6 MATCH (m)-[rm:MADE_TRANSACTION]->()
7 WITH m,n,sharedTerminals,COUNT(rm) AS txCountM
8 MATCH (n)-[rn:MADE_TRANSACTION]->()
9 WITH m,n,sharedTerminals,txCountM,COUNT(rn) AS txCountN
10 WHERE abs(txCountM - txCountN) <= 2
11 RETURN
12 m.id AS CustomerM,
13 n.id AS CustomerN,
14 sharedTerminals,
15 txCountM,
16 txCountN
```

Table RAW

	CustomerM	CustomerN	sharedTerminals	txCountM	txCountN
1	591	1317	4	1158	1158

Started streaming 1 record after 1 ms and completed after 1 hour 30 minutes.

50MB → 100MB

4.8 min → 18.6 min

≈ 3.9× slower

100MB → 200MB

18.6 min → 90 min

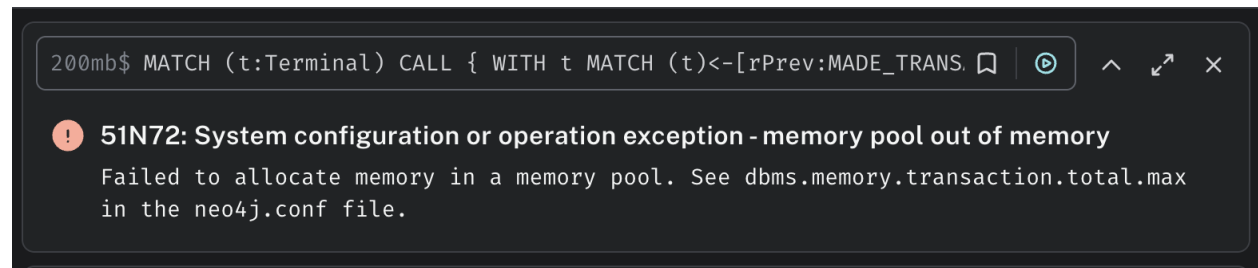
≈ 4.8× slower

Query A exhibited super-linear growth. While the dataset size increased by a factor of two between experiments, execution time increased by approximately 4–5×. At 200MB the

query required approximately 90 minutes to complete, demonstrating that pairwise customer comparison becomes increasingly expensive as graph density grows.

QUERY B .

Query B successfully executed on the 50MB dataset but exceeded the available transaction memory on both the 100MB and 200MB datasets. The query requires large intermediate collections and sorting operations, making it highly memory intensive and unsuitable for larger graph instances without additional optimization or memory allocation.

A screenshot of a Neo4j terminal window. The terminal has a dark background with light-colored text. At the top, there is a command prompt '200mb\$' followed by a Cypher query: 'MATCH (t:Terminal) CALL { WITH t MATCH (t)<-[rPrev:MADE_TRANS. The query requires large intermediate collections and sorting operations, making it highly memory intensive and unsuitable for larger graph instances without additional optimization or memory allocation.' Below the command prompt, there is an error message. It starts with an orange circle containing an exclamation mark, followed by the text '51N72: System configuration or operation exception - memory pool out of memory'. Below this, it says 'Failed to allocate memory in a memory pool. See dbms.memory.transaction.total.max in the neo4j.conf file.' The terminal window has a standard toolbar with icons for back, forward, search, and other navigation functions.

```
200mb$ MATCH (t:Terminal) CALL { WITH t MATCH (t)<-[rPrev:MADE_TRANS.
! 51N72: System configuration or operation exception - memory pool out of memory
Failed to allocate memory in a memory pool. See dbms.memory.transaction.total.max
in the neo4j.conf file.
```

QUERY C.

200mb\$ MATCH (u1:Customer {id: 0})-[:MADE_TRANSACTION]->(t1:Terminal

TableRAW

Q

T

↓

	u3.id
1	134
2	154
3	476
4	479
5	657
6	764
7	959
8	1018
9	1376
10	9

Started streaming 50 records after 278 ms and completed after 828 ms.

> ⓘ 03N96: Code generation failed

QUERY D 2

```
1 MATCH (c1:Customer)-[r1:MADE_TRANSACTION]->(t:Terminal)
2 MATCH (c2:Customer)-[r2:MADE_TRANSACTION]->(t)
3 WHERE c1.id < c2.id
4 WITH c1,c2,t,
5 count(r1) AS tx1,
6 count(r2) AS tx2,
7 avg(r1.satisfactionRating) AS avgSat1,
8 avg(r2.satisfactionRating) AS avgSat2
9 WHERE tx1 >= 5
10 AND tx2 >= 5
11 AND abs(avgSat1 - avgSat2) <= 0.5
12 MERGE (c1)-[:FREQUENT_COLLABORATOR {
13 terminalId: t.id,
14 avgRatingDiff: abs(avgSat1 - avgSat2)
15 }]->(c2)
16 RETURN count(*);
```

Table RAW

count(*)

1	33697
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✓ Created 33,697 relationships, set 67,394 properties
Started streaming 1 record after 2 ms and completed after 2 hours 47 minutes.

Increasing the dataset size did not significantly increase the number of discovered collaborator relationships, but it substantially increased the computational cost required to identify them.

QUERY E

- **Thursday** has the highest outlier percentage (~0.1400%)
- **Wednesday** has the lowest outlier percentage (~0.1238%)
- The percentages are remarkably stable across the week, staying around **0.12–0.14%**

200mb\$ MATCH ()-[r:MADE_TRANSACTION]->() WITH CASE datetime(replace(

Table RAW

Q T v

	dayOfWeek	totalTransactions	outlierTransactions	outlierPercentage
1	"Friday"	578082	805	0.13925360070024667
2	"Monday"	581064	786	0.13526909256123248
3	"Saturday"	579216	791	0.1365639070743902
4	"Sunday"	583908	789	0.13512402638771862
5	"Thursday"	578656	810	0.13997953879334182
6	"Tuesday"	582248	799	0.1372267487393688
7	"Wednesday"	578220	716	0.12382830064681263

Started streaming 7 records after 211 ms and completed after 40,578 ms.